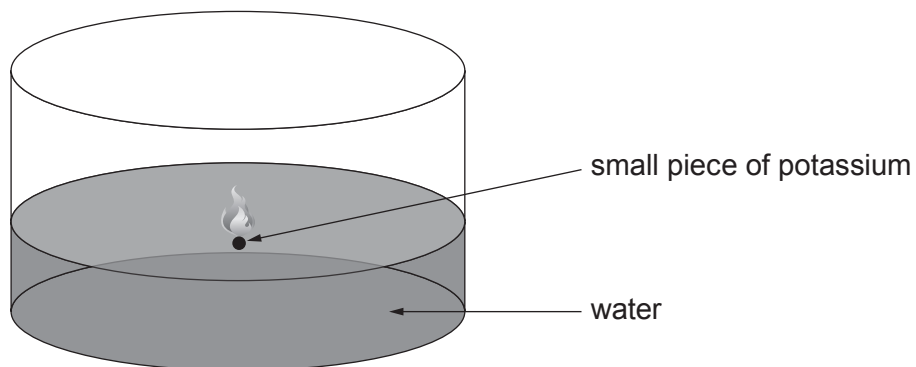


Answer **all** questions.

1. (a) Potassium reacts with water to produce potassium hydroxide and hydrogen gas.



- (i) Complete the table. Tick (✓) **one** box in **each** of the last two rows.

[2]

Substance	Formula	Element	Compound
potassium	K	✓	
water	H ₂ O		✓
potassium hydroxide	KOH		
hydrogen	H ₂		

- (ii) During the reaction, potassium burns with a **lilac** flame.

Give **two** other observations that you would expect to make.

Choose your answers from the box.

[2]

orange crystals form

fizzing

potassium sinks

potassium floats

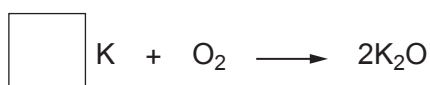
white precipitate forms

Observation 1

Observation 2



- (b) Potassium also reacts with oxygen to produce potassium oxide.



Choose a number from the box to balance the equation for this reaction.

[1]

2

4

6

- (c) Most Group 1 and Group 2 metals can be identified by the colour seen in a flame test.

Draw **one** line from each metal to the colour seen.

One line has been drawn for you.

[2]

Metal

Colour seen

lithium

white

sodium

red

barium

green

blue

yellow

